

Using the dimensionless variables that were defined in Eqs. (7.7.1) through (7.7.4), the linearized equations of motion that were given by Eqs. (7.5.23) and (7.5.24) may be written as

the **nondimensional linearized longitudinal equations**,

$$\begin{bmatrix} (1 - R_{x,\hat{\mu}}) & -R_{x,\hat{\alpha}} & 0 & 0 & 0 & 0 \\ -R_{z,\hat{\mu}} & (1 - R_{z,\hat{\alpha}}) & 0 & 0 & 0 & 0 \\ -R_{m,\hat{\mu}} & -R_{m,\hat{\alpha}} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\hat{\mu} \\ \Delta\hat{\alpha} \\ \Delta\hat{q} \\ \Delta\hat{\xi}_x \\ \Delta\hat{\xi}_z \\ \Delta\hat{\theta} \end{bmatrix} \quad (7.7.5)$$

$$= \begin{bmatrix} R_{x,\mu} & R_{x,\alpha} & R_{x,\bar{q}} & 0 & 0 & -R_{gx} \cos \theta_o \\ R_{z,\mu} & R_{z,\alpha} & (R_{z,\bar{q}} + 1) & 0 & 0 & -R_{gx} \sin \theta_o \\ R_{m,\mu} & R_{m,\alpha} & R_{m,\bar{q}} & 0 & 0 & 0 \\ \cos \theta_o & \sin \theta_o & 0 & 0 & 0 & -\sin \theta_o \\ -\sin \theta_o & \cos \theta_o & 0 & 0 & 0 & -\cos \theta_o \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\mu \\ \Delta\alpha \\ \Delta\bar{q} \\ \Delta\xi_x \\ \Delta\xi_z \\ \Delta\theta \end{bmatrix} + \begin{bmatrix} R_{x,\delta_e} \\ R_{z,\delta_e} \\ R_{m,\delta_e} \\ 0 \\ 0 \\ 0 \end{bmatrix} \Delta\delta_e$$

and the **nondimensional linearized lateral equations**,

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -\iota_{xz} & 0 & 0 & 0 \\ 0 & -\iota_{zx} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\hat{\beta} \\ \Delta\hat{p} \\ \Delta\hat{r} \\ \Delta\hat{\xi}_y \\ \Delta\hat{\phi} \\ \Delta\hat{\psi} \end{bmatrix} \quad (7.7.6)$$

$$= \begin{bmatrix} R_{y,\beta} & R_{y,\bar{p}} & (R_{y,\bar{r}} - 1) & 0 & R_{gy} \cos \theta_o & 0 \\ R_{\ell,\beta} & R_{\ell,\bar{p}} & R_{\ell,\bar{r}} & 0 & 0 & 0 \\ R_{n,\beta} & R_{n,\bar{p}} & R_{n,\bar{r}} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & \cos \theta_o \\ 0 & 1 & \tan \theta_o & 0 & 0 & 0 \\ 0 & 0 & \sec \theta_o & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\beta \\ \Delta\bar{p} \\ \Delta\bar{r} \\ \Delta\xi_y \\ \Delta\phi \\ \Delta\psi \end{bmatrix} + \begin{bmatrix} R_{y,\delta_a} & R_{y,\delta_r} \\ R_{\ell,\delta_a} & R_{\ell,\delta_r} \\ R_{n,\delta_a} & R_{n,\delta_r} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta_a \\ \Delta\delta_r \end{bmatrix}$$

where we are using the notation for both longitudinal and lateral motion,

$$\hat{f} \equiv \frac{\partial f}{\partial \tau_x} \text{ for longitudinal motion and } \hat{f} \equiv \frac{\partial f}{\partial \tau_y} \text{ for lateral motion} \quad (7.7.7)$$